

# 4. Expectation

# Learning Outcomes

- By the end of this week, you should be able to:

- Employ common discrete distributions:
  - *Geometric, Negative Binomial, & Poisson*
- Understand what the *expected value* of a RV represents
- Calculate expected values of discrete RVs
  - Calculate expected values of sums/functions of RVs
- Interpret the *mean* & *variance* of a distribution

# Geometric Distribution

- Consider *infinite* sequence of *independent Bernoulli trials* with the *same probability* of "success" $p$



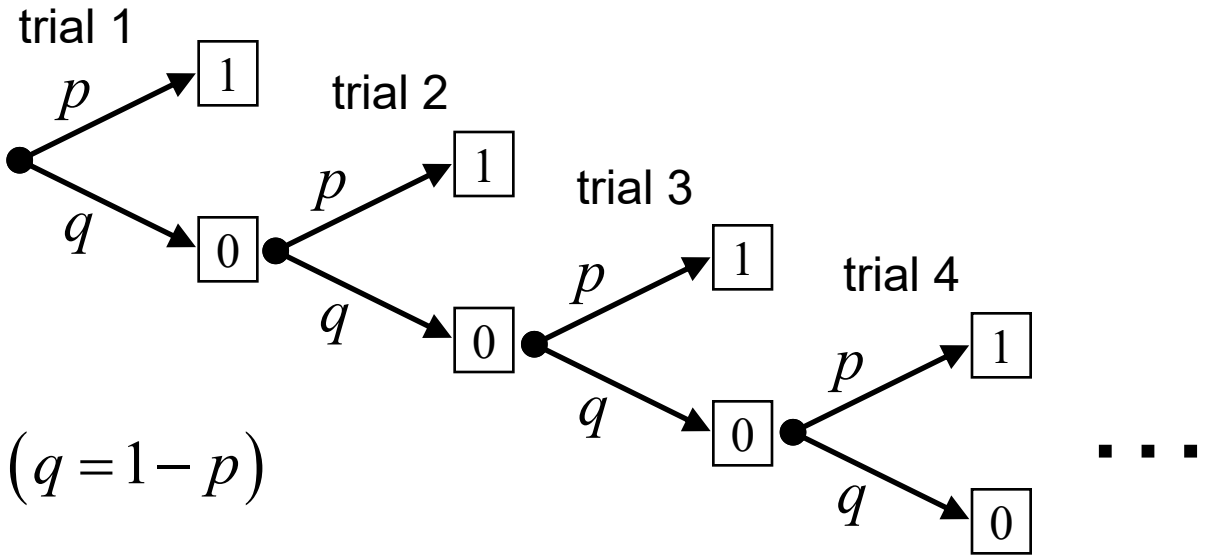

- Let RV $X/Y$ *count # of failures/trials until 1st success* $(Y = X + 1)$
  - E.g. $s = ($ 0 0 0 1 0 1 $\ldots\ ) \Rightarrow X(s) = 3 \,/\, Y(s) = 4$

- $X/Y$ called *Geometric* RV

# Geometric Distribution

- What are possible values of Geometric RV $X$?

- What is the probability of having $X$=3?




$(q = 1 - p)$

# Geometric Distribution

- More generally, PMF of Geometric RV is

$$P(X = x) = pq^x, \forall x \in \mathbb{N}_0$$

$$\Leftrightarrow P(Y = y) = pq^{y-1}, \forall y \in \mathbb{N}_+$$

  - Denoted $X \sim$ Geometric$(p)$
- Verify this is a valid PMF & find its CDF

Geometric Series:

$$\sum_{i=0}^{n} a^i = \frac{1-a^{n+1}}{1-a}$$

$$\sum_{i=0}^{\infty} a^i = \frac{1}{1-a}, \ a \in (0,1)$$

# Example



- You try to hit a target taking a sequence of shots. Assume the shots are independent with 5% probability of success.
  - What is the probability of hitting the target on the $5^{th}$ try?
  - What is the probability of not hitting the target in the first 10 shots?
  - What is the (conditional) probability of not hitting the target in the first 20 shots, given that you didn't hit it in the first 10 shots?

# Negative Binomial Distribution

- Consider *infinite* sequence of *independent Bernoulli trials* with the *same probability* of "success" $p$




- Let the RV $X/Y$ counts # of failures/trials until $r^{th}$ success
  - E.g. for $r$=3

outcome $s$ = ( 0 0 1 1 0 1 · · · ) $\Rightarrow$ $X(s)=3/Y(s)=6$

- Such $X$ is called *Negative Binomial* RV

# Negative Binomial Distribution

- For arbitrary $r \in \mathbb{N}$ , what are possible values of NegBin RV $X$ ?

- What is the probability of having $x=3$ failures for $r=3$ successes?

$$\{X = 5\} = \left\{ \begin{array}{cccccc} \boxed{1} & \boxed{1} & \boxed{0} & \boxed{0} & \boxed{0} & \boxed{1} \\ \boxed{1} & \boxed{0} & \boxed{1} & \boxed{0} & \boxed{0} & \boxed{1} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \boxed{0} & \boxed{0} & \boxed{0} & \boxed{1} & \boxed{1} & \boxed{1} \end{array} \right.$$

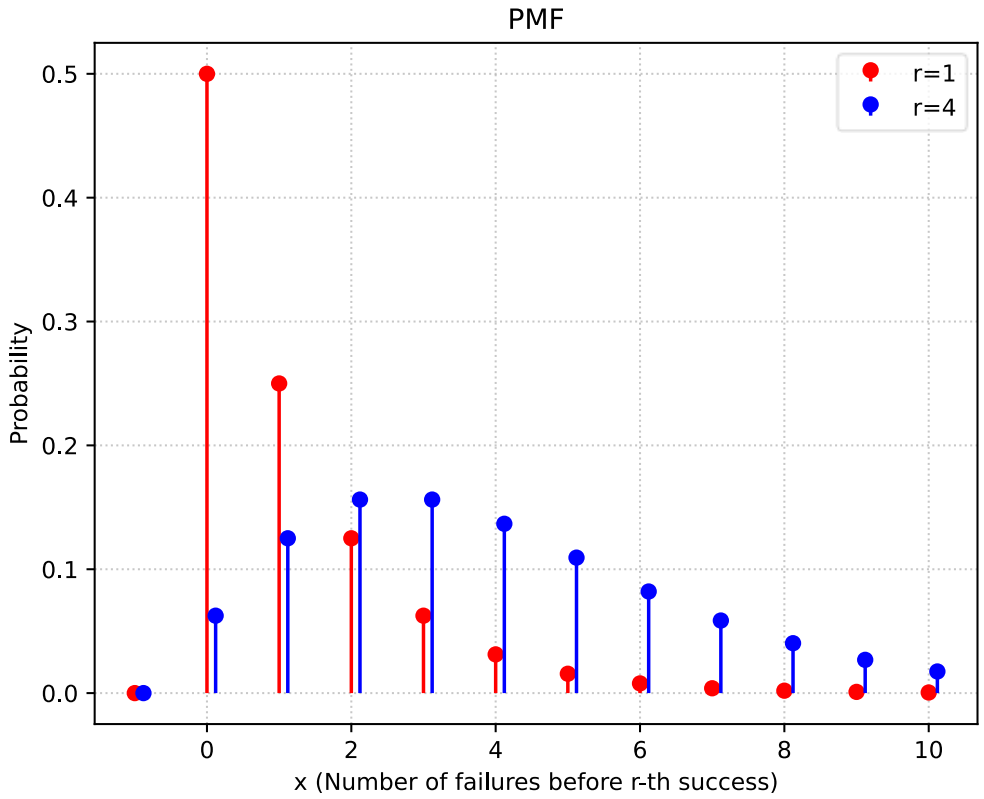
# Negative Binomial Distribution

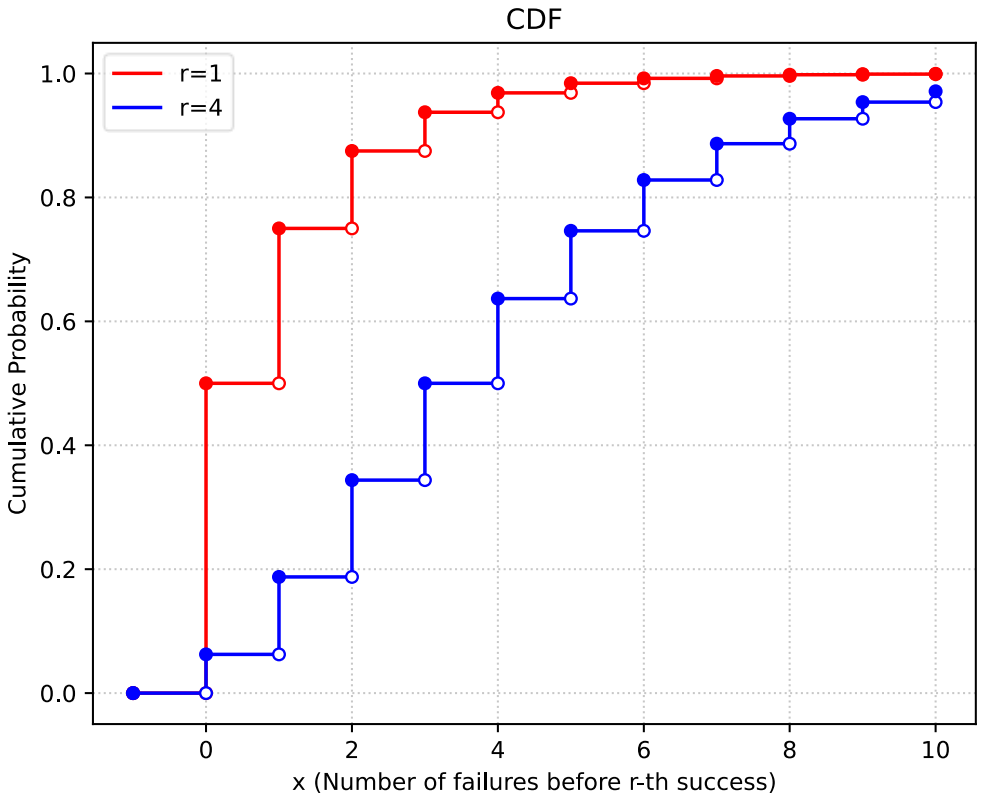
- More generally, PMF of Negative Binomial distribution is

$$p_X(x) = \binom{x+r-1}{r-1} p^r q^x, \ \forall x \in \mathbb{N}_0$$

  - If RV $Y$ counts # *trials* until $r^{\text{th}}$ success $\rightarrow Y = X + r$
  - Denoted $X \sim$ NegBinom$(r,p)$
- Negative Binomial is *sum* of identical & independent Geometrics
  - # trials till $r^{\text{th}}$ successes = (# trials till 1$^{\text{st}}$ ) + … + (# trials till $r^{\text{th}}$ )
  - Geometric is special case ($r=1$) of Negative Binomial

# Negative Binomial Distribution




same $p$=0.5

$r$=1

$r$=3

# Expected Value

- RV $X$ represents quantity of interest in random experiment, and its distribution measures uncertainty around value of $X$
- Often want to describe *aggregate* behavior of $X$ w/ single value
  - E.g., long-run proportion of Heads in sequence of coin flips, or average completion time for web-server requests
- *Expected Value* of RV $X$ is value we expect *on average,* over *many repetitions* of the random experiment

# Expected Value – Discrete RV

- Consider discrete RV $X$, and assume you repeat experiment #$n$ times, getting values $(c_1, ..., c_n)$; a*verage* of #$n$ values is

$$\frac{1}{n}\sum_i c_i = \sum_x x \times \overbrace{\left(\frac{(\#\text{ times } c_i = x)}{n}\right)}^{\approx p_X(x)} \approx \sum_x x \times p_X(x)$$

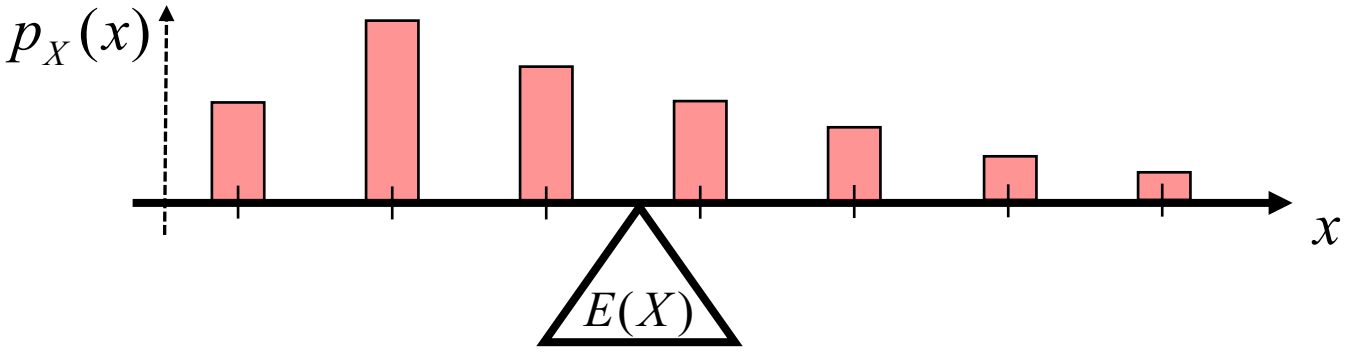
- For *discrete* RV $X$ with PMF $p_X(x)$, expected value is *defined* as

$$\boxed{E(X) = \sum_x x p_X(x)}$$

(provided sum converges!!)

# Expected Value

- Expected value (a.k.a. *expectation*/*mean*) of RV $X$ is *center of gravity* (weighted center) of distribution




  - $E(X)$ doesn't have to be one of possible $X$ values

# Example

- For sequence of independent Bernoulli($p$) trials, find the expected # of trials until the first success.
  - I.e., the expected value of the Geometric($p$) distribution

# Expected Value – Functions of RVs

- For discrete RV $X$, expected value of function $Y=g(X)$ is

$$E(Y) = \sum_y y p_Y(y) = \sum_x g(x) p_X(x) = E(g(X))$$

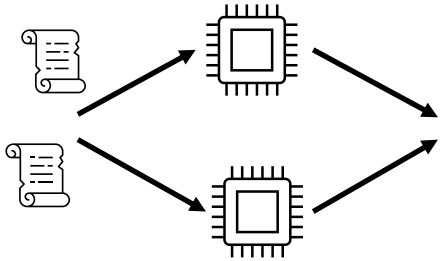
  - No need to derive distribution of $Y$, can just use PMF of $X$!
- Similarly for multivariate function $Z=g(X,Y)$ of discrete RVs $X,Y$

$$E[Z] = E[g(X,Y)] = \sum_x \sum_y g(x,y) P(X=x, Y=y)$$

- *Linearity* of expectations: for RVs $X,Y$

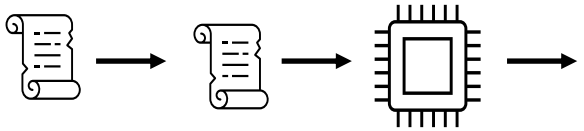
$$E(aX+bY) = aE(X) + bE(Y)$$

# Example



- Two processes run *in parallel* on two computers and take independently Geometric*(.5) times to complete. (* # trials) Find the average time until *both processes* complete?

# Example



- Two processes run *serially* (i.e., one after the other) on the same computer and take independent Geometric*(.5) times to complete. Find the average time until both processes complete.

# Expected Value – Indicator RVs

- Consider *indicator* RV $I_A(s) = \begin{cases} 1 & s \in A \\ 0 & s \notin A \end{cases}$, for some $A \subseteq S$

- Expected value of indicator RV is equal to probability of its defining event: $\boxed{E(I_A) = P(A)}$

  - Proof:

# Example (Matching problem)

- #$n$ people throw their keys in a box, and each one picks a random set of keys. Find the expected number of people who get back their own keys?

# Variance

- *Variance* is measure of spread defined as *expected squared distance* of RV from its mean $E(X)$

$$V(X) = E\left((X - \underbrace{E(X)}_{=\mu})^2\right) = E(X^2) - \mu^2$$

  - Variance typically denoted by $\sigma^2$
  - *Standard Deviation* (*SD*) is *square root* of variance $\sigma = \sqrt{\sigma^2}$

# Example

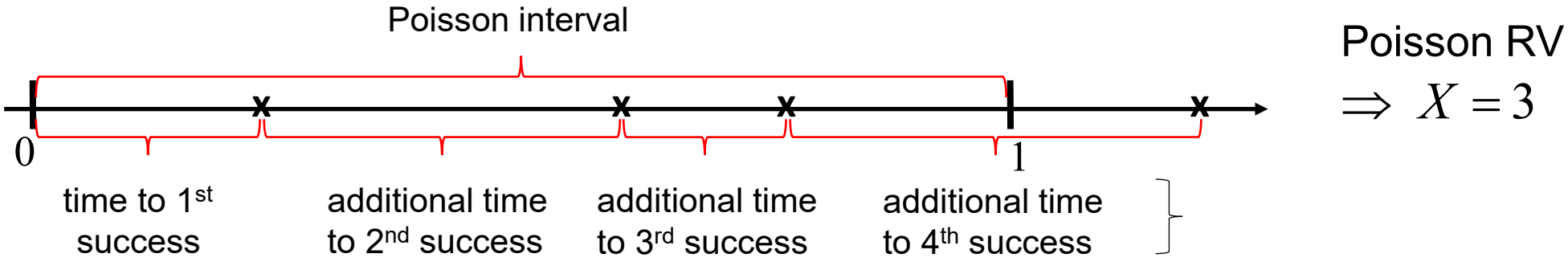
- Show that $E\left((X \ -\mu)^2\right) = E(X^2) - \mu^2$, where $\mu = E(X)$

# Example

- Find the variance of the Geometric($p$) distribution

# Poisson Distribution

- Want distribution of # goals in a hockey game
  - If # attempts in game was known → use Binomial
  - However, do not want to restrict # attempts/goals beforehand
- A *Poisson* RV counts # of successes in some *continuous interval* (e.g. time, distance, area or volume)
  - If time to between successes is random and memoryless, then Poisson RV counts # of successes within a given interval



# Poisson Distribution

Exponentials:

$$e^x = \sum_{i=0}^{\infty} \frac{x^i}{i!}$$

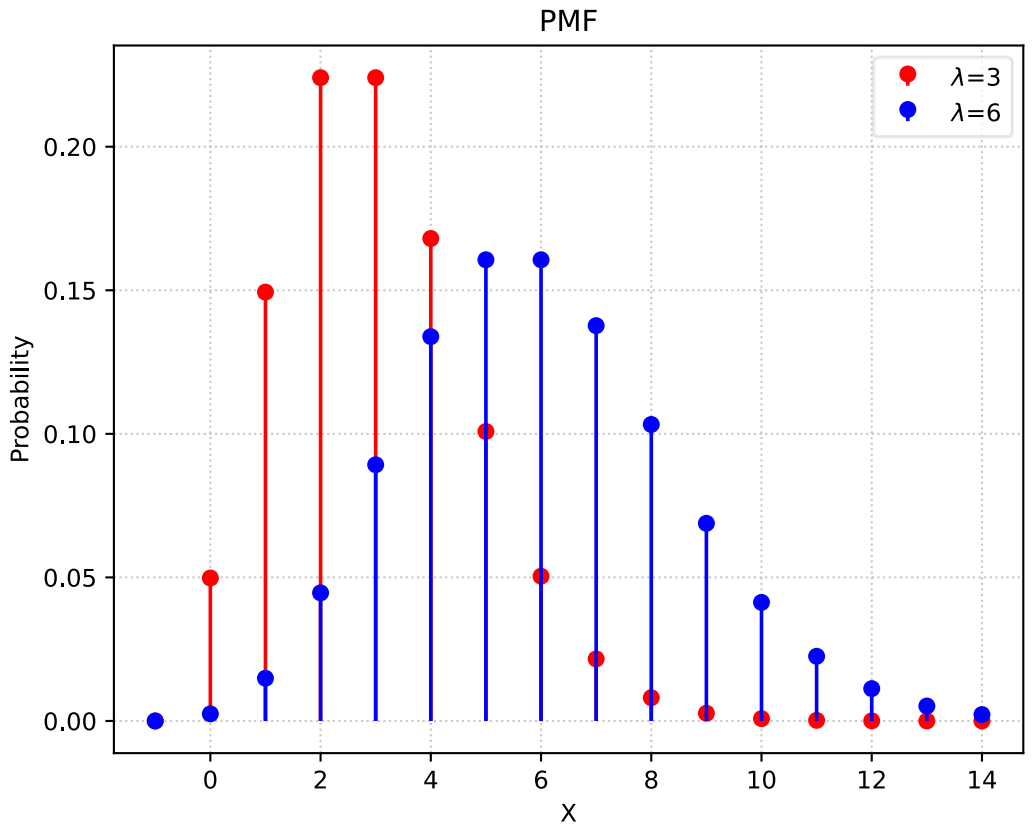
$$= 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \ldots$$

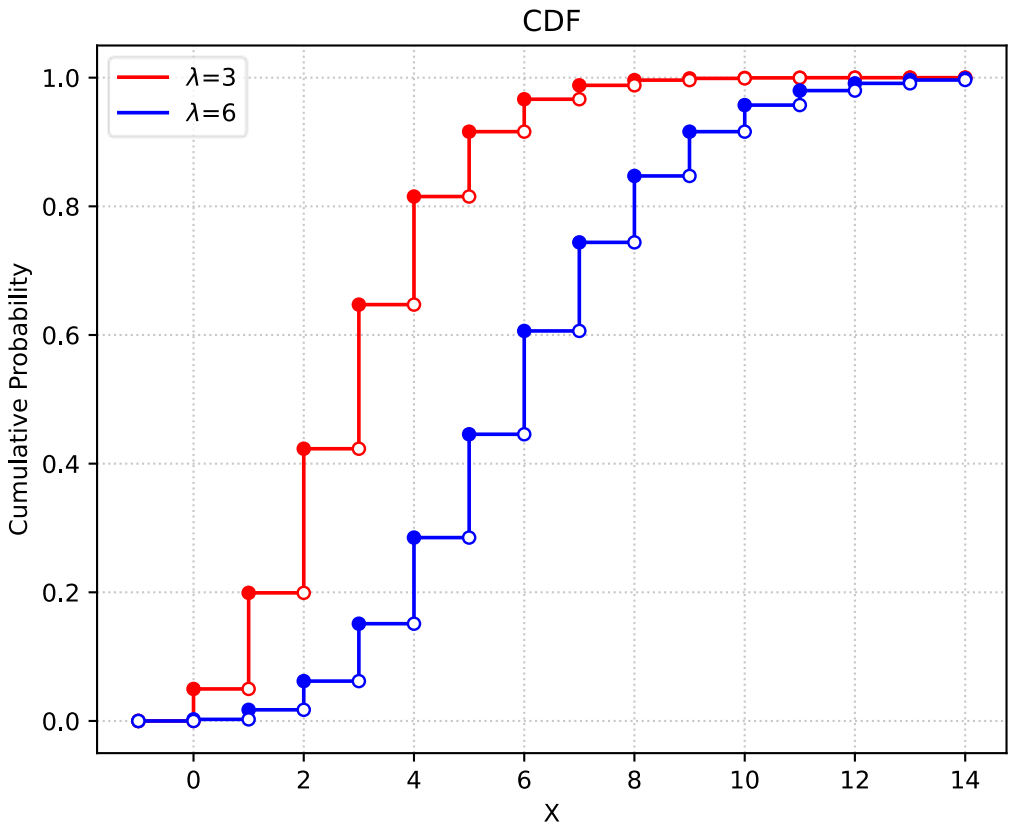
- PMF of Poisson distribution is

$$P(X = x) = \frac{\lambda^x}{x!} e^{-\lambda}, \quad x = 0, 1, 2, \ldots$$

  - Parameter $\lambda > 0$ represents average # of successes over interval
  - Denoted $X \sim \text{Poisson}(\lambda)$
- Verify this is a valid PMF

# Poisson Distribution



# Poisson Approximation to Binomial

- Poisson($\lambda$) is *approximated* by Binomial($n$,$\lambda/n$) as $n \to \infty$
  - Think of dividing Poisson interval to $n$ pieces, each with $p=\lambda/n$ probability of containing *single* success

